Лаб 2. Docker. 353502 Бурчук Дмитрий вариант 5.

1. Подготовьте рабочее окружение в соответствии с типом вашей операционной системы

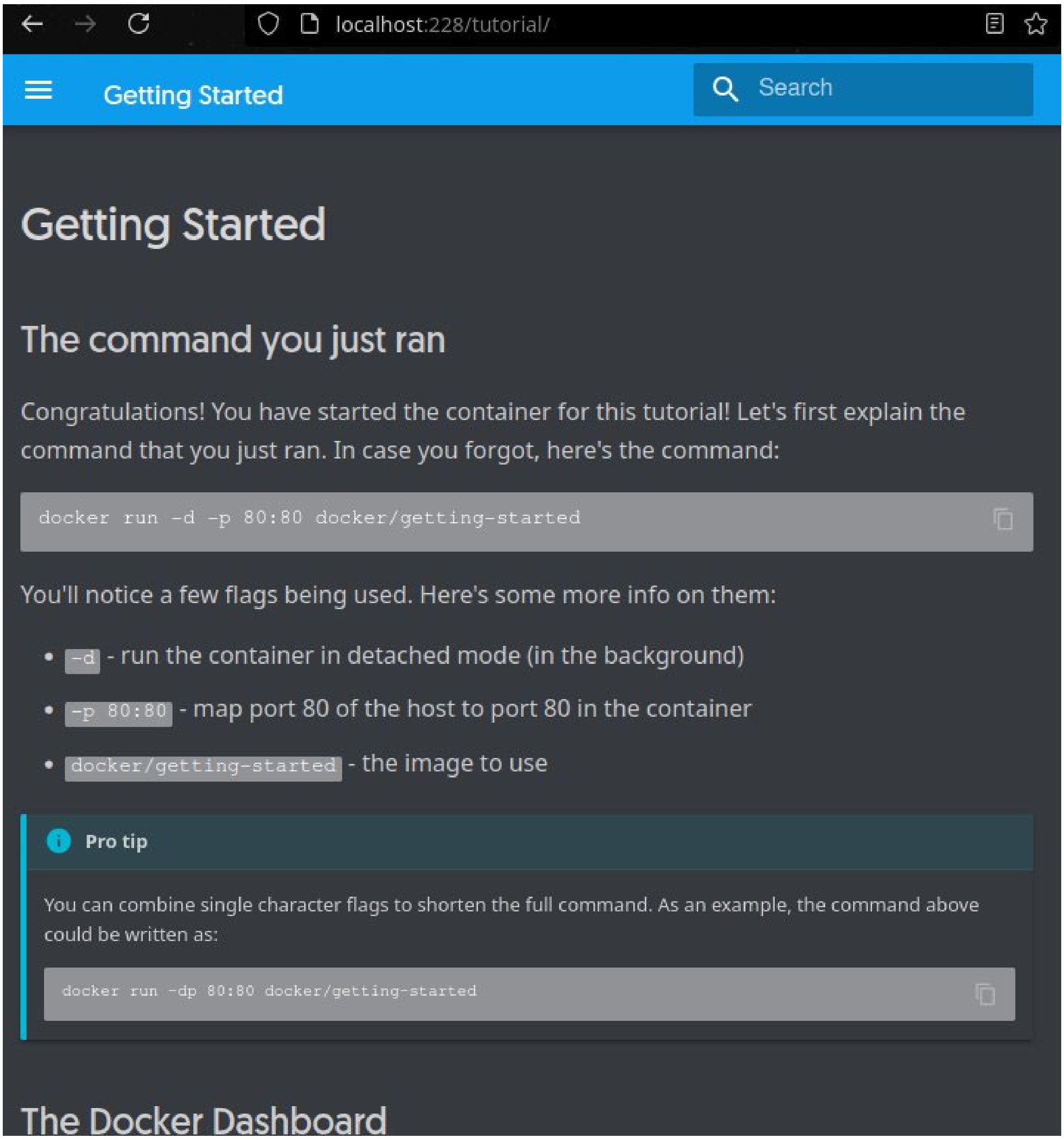
```
~ docker -v
Docker version 27.5.1, build 9f9e405801
```

2. Изучите простейшие консольные команды и возможности Docker Desktop (см. лекцию), создать собственный контейнер docker/getting-started, открыть в браузере и изучить tutorial

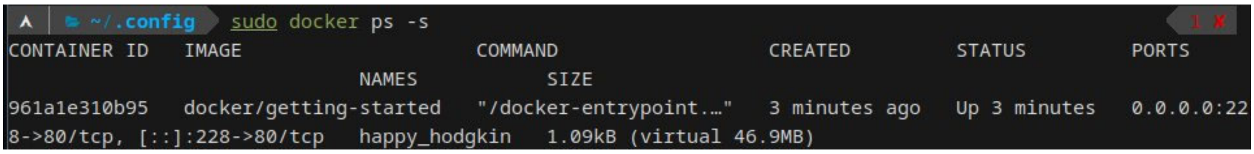
```
~ sudo usermod -aG docker G1nxx
```

Запуск докер контейнера getting-started

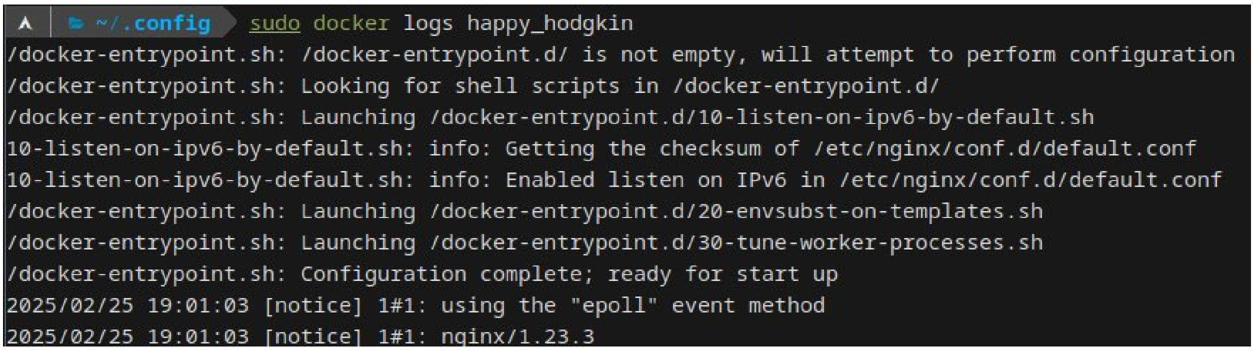
```
~/.config sudo docker run -d -p 228:80 docker/getting-started
Unable to find image 'docker/getting-started:latest' locally
latest: Pulling from docker/getting-started
c158987b0551: Pull complete
1e35f6679fab: Pull complete
cb9626c74200: Pull complete
b6334b6ace34: Pull complete
f1d1c9928c82: Pull complete
9b6f639ec6ea: Pull complete
ee68d3549ec8: Pull complete
33e0cbbb4673: Pull complete
4f7e34c2de10: Pull complete
Digest: sha256:d79336f4812b6547a53e735480dde67f8f8f7071b414fbd9297609ffb989abc1
Status: Downloaded newer image for docker/getting-started:latest
961a1e310b95936be3e4c48256401776560dd5b8978febfcb9420c6ec293b36a
```



Вывод запущенных контейнеров с указанием их размеров



Вывод логов контейнера



Остановка контейнера

```
~/.config  sudo docker stop happy_hodgkin
happy_hodgkin
~/.config  sudo docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
~/.config  sudo docker ps -a
CONTAINER ID   IMAGE                    COMMAND                  CREATED         STATUS
   PORTS     NAMES
961a1e310b95   docker/getting-started   "/docker-entrypoint.…"   8 minutes ago   Exited (0) 18 seconds ago
             happy_hodgkin
```

Запуск и приостановка контейнера

```
~/.config  sudo docker ps
CONTAINER ID   IMAGE                    COMMAND                  CREATED              STATUS
PORTS                                       NAMES
b31a4ce18120   docker/getting-started   "/docker-entrypoint.…"   About a minute ago   Up About a minute
0.0.0.0:228->80/tcp, [::]:228->80/tcp   suspicious_napier
~/.config  sudo docker pause suspicious_napier
suspicious_napier
~/.config  sudo docker ps
CONTAINER ID   IMAGE                    COMMAND                  CREATED              STATUS
         PORTS                                   NAMES
b31a4ce18120   docker/getting-started   "/docker-entrypoint.…"   About a minute ago   Up About a minute (P
aused)   0.0.0.0:228->80/tcp, [::]:228->80/tcp   suspicious_napier
```

Удаление контейнера

```
~/.config  sudo docker stop suspicious_napier
suspicious_napier
~/.config  sudo docker rm suspicious_napier
suspicious_napier
~/.config  sudo docker ps -a
CONTAINER ID   IMAGE                    COMMAND                  CREATED          STATUS
   PORTS     NAMES
961a1e310b95   docker/getting-started   "/docker-entrypoint.…"   15 minutes ago   Exited (0) 7 minutes ago
             happy_hodgkin
```

Удаление образа

```
~/.config  sudo docker images
REPOSITORY               TAG       IMAGE ID       CREATED         SIZE
docker/getting-started   latest    3e4394f6b72f   2 years ago     46.9MB
~/.config  sudo docker image rm -f 3e43
Untagged: docker/getting-started:latest
Untagged: docker/getting-started@sha256:d79336f4812b6547a53e735480dde67f8f8f7071b414fbd9297609ffb989abc1
Deleted: sha256:3e4394f6b72fccefa2217067a7f7ff84d5d828afa9623867d68fce4f9d862b6c
~/.config  sudo docker images
REPOSITORY   TAG       IMAGE ID   CREATED   SIZE
```

3. Создайте docker image, который запускает скрипт с использованием функций из https://github.com/smartiqaorg/geometric_lib.
    a. Данные необходимые для работы скрипта передайте любым удобным способом (например: конфиг файл через docker volume, переменные окружения, перенаправление ввода). Изучите простейшие консольные команды для работы с docker(см. лекцию). Зарегистрируйтесь на DockerHub и выберите необходимые для проекта образы

b. Создать Dockerfile для реализации сборки собственных Docker образов
c. Использовать его для создания контейнера. Протестировать использование контейнера

Dockerfile:

```
~/Projects/ABC/LR1  cat dockerfile

File: dockerfile

FROM python:3

WORKDIR /usr/src/app

COPY ./script.py .

VOLUME /usr/src/app/files

ENTRYPOINT ["python", "./script.py"]
```

Build:

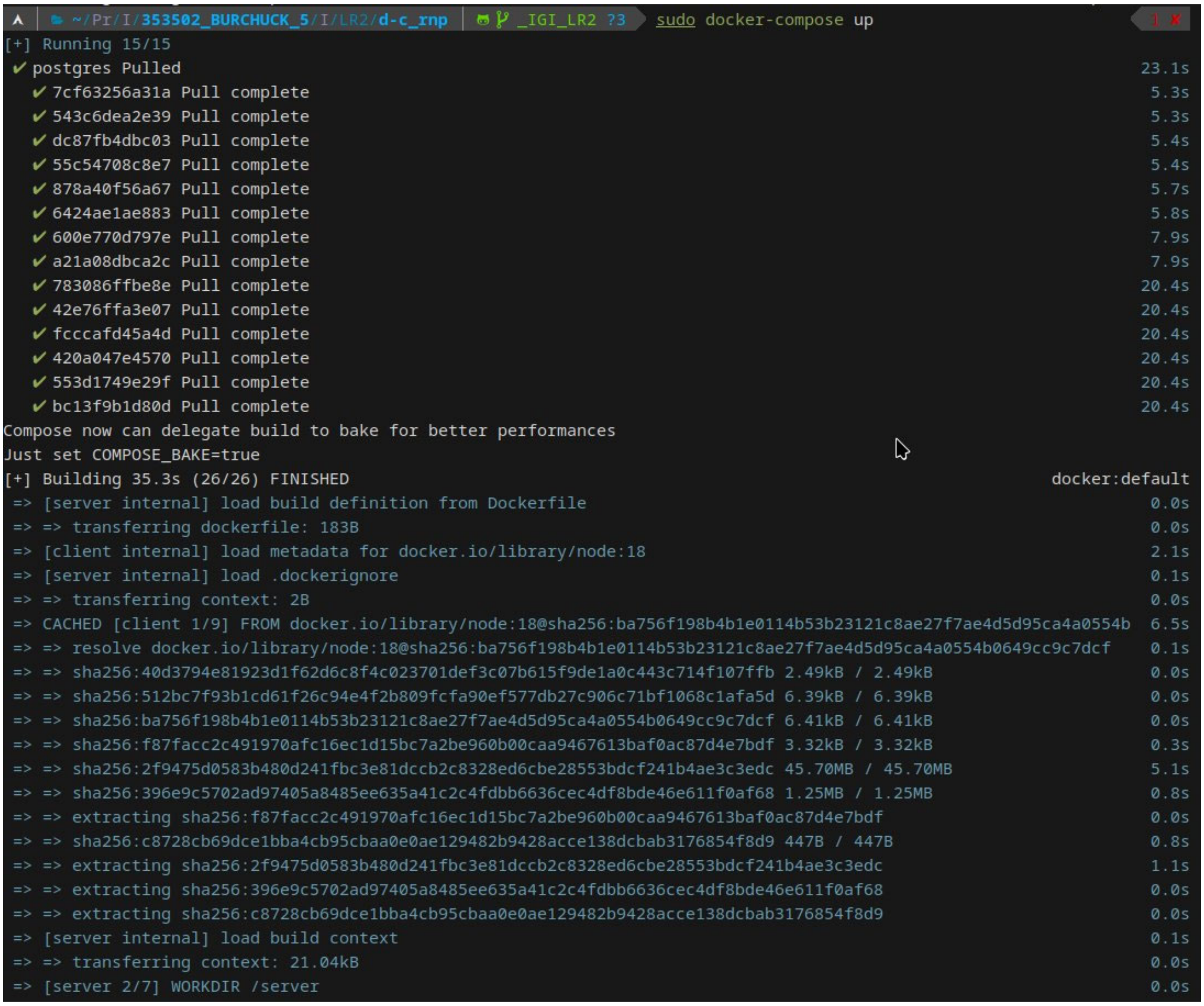
```
~/Pr/I/353502_BURCHUCK_5/I/LR1   _IGI_LR1 ?2  sudo docker build . -t py_o:1
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  17.41kB
Step 1/5 : FROM python:3
 ---> 36d17f72f4f3
Step 2/5 : WORKDIR /usr/src/app
 ---> Using cache
 ---> 8b13fbb65f19
Step 3/5 : COPY ./script.py .
 ---> Using cache
 ---> f58ec7fab664
Step 4/5 : VOLUME /usr/src/app/files
 ---> Using cache
 ---> 423d68f6a89c
Step 5/5 : ENTRYPOINT ["python", "./script.py"]
 ---> Using cache
 ---> 52946af3e945
Successfully built 52946af3e945
Successfully tagged py_o:1
```

Run:

```
~/Pr/I/353502_BURCHUCK_5/I/LR1   _IGI_LR1 ?2  sudo docker run -v ./:/usr/src/app/files -i py_o
:1
result :  5615813.638184853
```

4. Скачать любой доступный проект с GitHub с произвольным стеком технологий (пример – см. индивидуальное задание) или использовать свой, ранее разработанный. Создать для него необходимый контейнер, используя Docker Compose для управления многоконтейнерными приложениями. Запустить проект в контейнере.

Сборка контейнеров и запуск



```
~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp  _IGI_LR2 ?3  sudo docker-compose up
[+] Running 15/15
 ✔ postgres Pulled                                                          23.1s
   ✔ 7cf63256a31a Pull complete                                             5.3s
   ✔ 543c6dea2e39 Pull complete                                             5.3s
   ✔ dc87fb4dbc03 Pull complete                                             5.4s
   ✔ 55c54708c8e7 Pull complete                                             5.4s
   ✔ 878a40f56a67 Pull complete                                             5.7s
   ✔ 6424ae1ae883 Pull complete                                             5.8s
   ✔ 600e770d797e Pull complete                                             7.9s
   ✔ a21a08dbca2c Pull complete                                             7.9s
   ✔ 783086ffbe8e Pull complete                                             20.4s
   ✔ 42e76ffa3e07 Pull complete                                             20.4s
   ✔ fcccafd45a4d Pull complete                                             20.4s
   ✔ 420a047e4570 Pull complete                                             20.4s
   ✔ 553d1749e29f Pull complete                                             20.4s
   ✔ bc13f9b1d80d Pull complete                                             20.4s
Compose now can delegate build to bake for better performances
Just set COMPOSE_BAKE=true
[+] Building 35.3s (26/26) FINISHED                                 docker:default
 => [server internal] load build definition from Dockerfile                 0.0s
 => => transferring dockerfile: 183B                                        0.0s
 => [client internal] load metadata for docker.io/library/node:18           2.1s
 => [server internal] load .dockerignore                                    0.1s
 => => transferring context: 2B                                             0.0s
 => CACHED [client 1/9] FROM docker.io/library/node:18@sha256:ba756f198b4b1e0114b53b23121c8ae27f7ae4d5d95ca4a0554b  6.5s
 => => resolve docker.io/library/node:18@sha256:ba756f198b4b1e0114b53b23121c8ae27f7ae4d5d95ca4a0554b0649cc9c7dcf  0.1s
 => => sha256:40d3794e81923d1f62d6c8f4c023701def3c07b615f9de1a0c443c714f107ffb 2.49kB / 2.49kB  0.0s
 => => sha256:512bc7f93b1cd61f26c94e4f2b809fcfa90ef577db27c906c71bf1068c1afa5d 6.39kB / 6.39kB  0.0s
 => => sha256:ba756f198b4b1e0114b53b23121c8ae27f7ae4d5d95ca4a0554b0649cc9c7dcf 6.41kB / 6.41kB  0.0s
 => => sha256:f87facc2c491970afc16ec1d15bc7a2be960b00caa9467613baf0ac87d4e7bdf 3.32kB / 3.32kB  0.3s
 => => sha256:2f9475d0583b480d241fbc3e81dccb2c8328ed6cbe28553bdcf241b4ae3c3edc 45.70MB / 45.70MB  5.1s
 => => sha256:396e9c5702ad97405a8485ee635a41c2c4fdbb6636cec4df8bde46e611f0af68 1.25MB / 1.25MB  0.8s
 => => extracting sha256:f87facc2c491970afc16ec1d15bc7a2be960b00caa9467613baf0ac87d4e7bdf  0.0s
 => => sha256:c8728cb69dce1bba4cb95cbaa0e0ae129482b9428acce138dcbab3176854f8d9 447B / 447B  0.8s
 => => extracting sha256:2f9475d0583b480d241fbc3e81dccb2c8328ed6cbe28553bdcf241b4ae3c3edc  1.1s
 => => extracting sha256:396e9c5702ad97405a8485ee635a41c2c4fdbb6636cec4df8bde46e611f0af68  0.0s
 => => extracting sha256:c8728cb69dce1bba4cb95cbaa0e0ae129482b9428acce138dcbab3176854f8d9  0.0s
 => [server internal] load build context                                    0.1s
 => => transferring context: 21.04kB                                        0.0s
 => [server 2/7] WORKDIR /server                                            0.0s
```

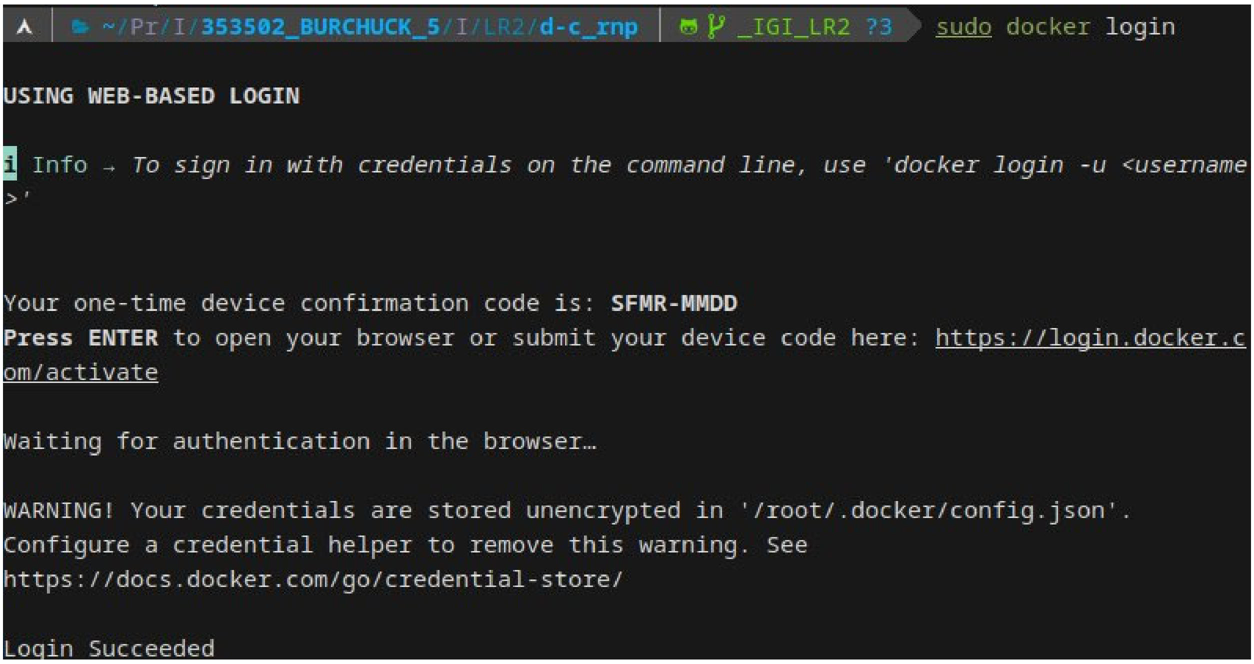
```
client      |   ➜  Local:   http://localhost:4173/
client      |   ➜  Network: http://172.18.0.4:4173/
server      | Prisma schema loaded from prisma/schema.prisma
server      | Datasource "db": PostgreSQL database "docker_test_db", schema "public" at "postgres:5432"
server      |
database    | 2025-03-03 18:13:27.326 UTC [79] ERROR:  table "_prisma_migrations" does not exist
database    | 2025-03-03 18:13:27.326 UTC [79] STATEMENT:  DROP TABLE _prisma_migrations
server      | Applying migration `20240526103908_`
server      | Applying migration `20240526104232_`
server      | Applying migration `20240526110749_`
server      |
server      | Database reset successful
server      |
server      | The following migration(s) have been applied:
server      |
server      | migrations/
server      |   └─ 20240526103908_/
server      |     └─ migration.sql
server      |   └─ 20240526104232_/
server      |     └─ migration.sql
server      |   └─ 20240526110749_/
server      |     └─ migration.sql
server      |
✔ Generated Prisma Client (v5.22.0) to ./node_modules/@prisma/client in 27ms
server      |
server      | Running seed command `node ./prisma/seed.js` ...
server      | Database is successully seeded
server      |
server      | 🌱  The seed command has been executed.
server      |
server      |
server      | > crud-server-express@1.0.0 start
server      | > node src/app.js
server      |
server      | Listening to port: 8000
server      | 2025-03-03T18:14:49.909Z - All users request hit!
server      | 2025-03-03T18:14:57.241Z - All users request hit!
server      | 2025-03-03T18:14:58.071Z - All users request hit!
```

5. Настроить сети и тома для обеспечения связи между контейнерами и сохранения данных (исходные данные, логин, пароль и т.д.)

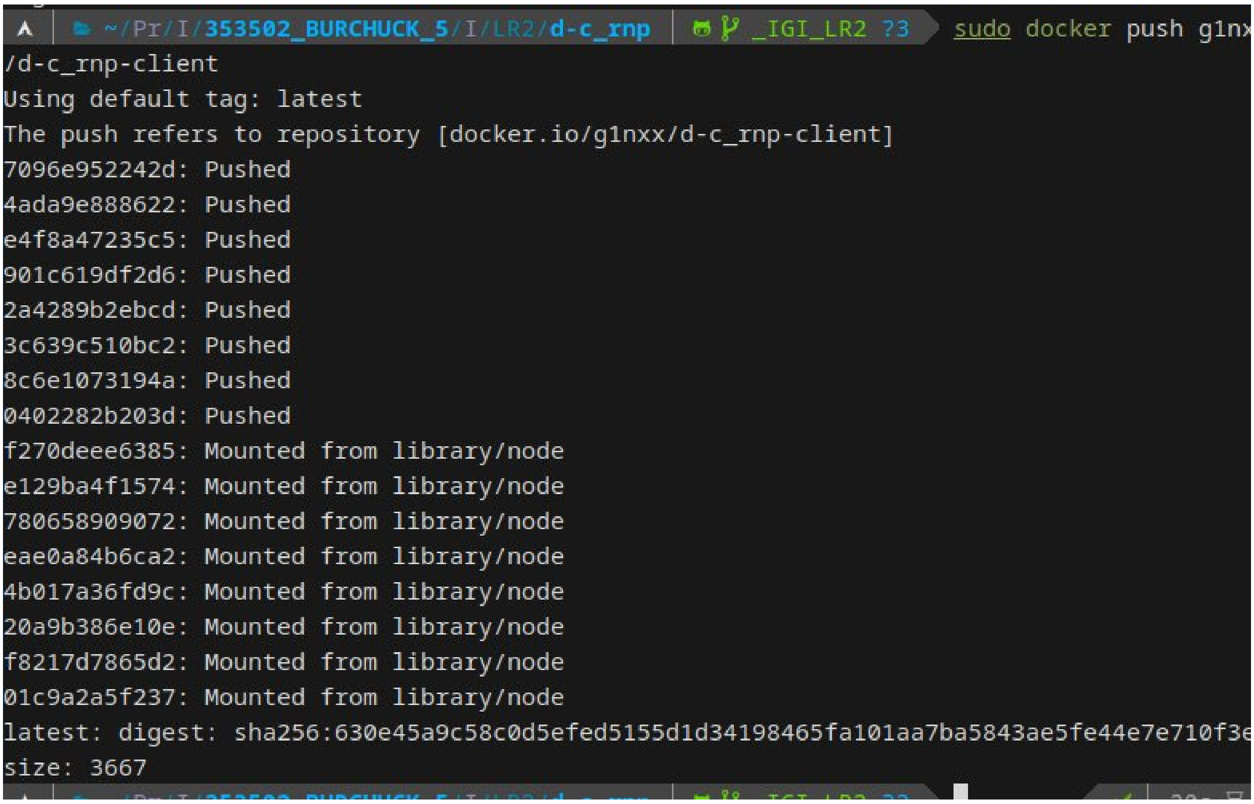
Docker-compose файл с добавочными данными

```yaml
    server:
        container_name: server
        build:
            context: ./server
            dockerfile: Dockerfile
        ports:
            - "7999:8000"
        command: bash -c "npx prisma migrate reset --force && npm start"
        environment:
            DATABASE_URL: "${DATABASE_URL}"
            PORT: "${SERVER_PORT}"
        depends_on:
            postgres:
                condition: service_healthy
    client:
        container_name: client
        build:
            context: ./client
            dockerfile: Dockerfile
        command: bash -c "npm run preview"
        ports:
            - "4172:4173"
        depends_on:
            - server

volumes:
  my_data_volume:

networks:
  my_network:
```

6. Разместите результат в созданный репозиторий в DockerHub



Закидывание имейджа на Docker Hub



```
 ~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp   _IGI_LR2 ?3  sudo docker push g1nxx
/d-c_rnp-server
Using default tag: latest
The push refers to repository [docker.io/g1nxx/d-c_rnp-server]
ab616c7e0d4d: Pushed
01763d5f2891: Pushed
79839e39e7ed: Pushed
8eb15488d9d8: Pushed
aeeb24105f17: Pushed
c891b863cda1: Pushed
f270deee6385: Mounted from g1nxx/d-c_rnp-client
e129ba4f1574: Mounted from g1nxx/d-c_rnp-client
780658909072: Mounted from g1nxx/d-c_rnp-client
eae0a84b6ca2: Mounted from g1nxx/d-c_rnp-client
4b017a36fd9c: Mounted from g1nxx/d-c_rnp-client
20a9b386e10e: Mounted from g1nxx/d-c_rnp-client
f8217d7865d2: Mounted from g1nxx/d-c_rnp-client
01c9a2a5f237: Mounted from g1nxx/d-c_rnp-client
latest: digest: sha256:79347a0fd67b6c9b3c002cbb2e3c72bfd1fa06ba49fa64b93677830582871920
size: 3256
```

7. Выполните следующие действия с целью изучить особенности сетевого взаимодействия:
   - Получить информацию о всех сетях, работающих на текущем хосте и подробности о каждом типе сети
   - Создать свою собственную сеть bridge, проверить, создана ли она, запустить Docker-контейнер в созданной сети, вывести о ней всю информацию(включая IP-адрес контейнера), отключить сеть от контейнера
   - Создать еще одну сеть bridge, вывести о ней всю информацию, запустить в ней три контейнера, подключиться к любому из контейнеров и пропинговать два других из оболочки контейнера, убедиться, что между контейнерами происходит общение по IP-адресу
   - Создать свою собственную сеть overlay, проверить, создана ли она, вывести о ней всю информацию
   - Создать еще одну сеть overlay, проверить, создана ли она, вывести о ней всю информацию, удалить сеть
   - Попробовать создать сеть host, сохранить результат в отчет

Сведения о всех сетях:

```
 ~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp   _IGI_LR2 ?3  sudo docker network ls
NETWORK ID     NAME             DRIVER    SCOPE
654e177b667b   bridge           bridge    local
1b5af8d58e23   d-c_rnp_default  bridge    local
631cde74fdd9   host             host      local
e76b2335ed9c   none             null      local
```

Подробная информация

```
~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp  _IGI_LR2 ?3  sudo docker network inspect 1b5
[
    {
        "Name": "d-c_rnp_default",
        "Id": "1b5af8d58e23b91e8b68989ae16a843853fcfa341706c74def0761f060597c4e",
        "Created": "2025-03-03T21:13:21.072260861+03:00",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.18.0.0/16",
                    "Gateway": "172.18.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": {},
        "Options": {},
        "Labels": {
            "com.docker.compose.config-hash": "8bcecc3dfa7a41a2ab4dc877274fce1d0ba6e2cf141473fb021dad9cfbb54aec",
            "com.docker.compose.network": "default",
            "com.docker.compose.project": "d-c_rnp",
            "com.docker.compose.version": "2.33.1"
        }
    }
]
```

Создадим свою сеть типа bridge

```
~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp  _IGI_LR2 ?3  sudo docker network create -d bridge newbridgeNetwork
302fef477bbe2dbfe418add6c8ec35dace690ef2d077034e6ba04165e1b7307d
~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp  _IGI_LR2 ?3  sudo docker network ls
NETWORK ID     NAME               DRIVER    SCOPE
654e177b667b   bridge             bridge    local
1b5af8d58e23   d-c_rnp_default    bridge    local
631cde74fdd9   host               host      local
302fef477bbe   newbridgeNetwork   bridge    local
76b2335ed9c    none               null      local
```

```
 ~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp   _IGI_LR2 ?3  sudo docker inspect 302
```


```
[
    {
        "Name": "newbridgeNetwork",
        "Id": "302fef477bbe2dbfe418add6c8ec35dace690ef2d077034e6ba04165e1b7307d",
        "Created": "2025-03-03T22:02:53.179274974+03:00",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.19.0.0/16",
                    "Gateway": "172.19.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": {},
        "Options": {},
        "Labels": {}
    }
]
```


Подключим контейнер к сети

```
 ~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp   _IGI_LR2 ?3  sudo docker network connect newBridgeNetwork 27a6
 ~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp   _IGI_LR2 ?3  sudo docker network connect newBridgeNetwork fcb6
```

Отключим его

```
 ~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp   _IGI_LR2 ?3  sudo docker network disconnect newBridgeNetwork fcb6
```

Создадим новую сеть и подключим контейнеры

```
 ~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp   _IGI_LR2 ?3  sudo docker network connect Grohelbecker  27
 ~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp   _IGI_LR2 ?3  sudo docker network connect Grohelbecker  fc
 ~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp   _IGI_LR2 ?3  sudo docker network connect Grohelbecker  2e
```

Можем увидеть подключенные контейнеры

```
~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp  _IGI_LR2 ?3  sudo docker inspect Grohelbecker
[
    {
        "Name": "Grohelbecker",
        "Id": "8c9706f6b8870076852502b17523c106af6914a280a17ccd5ef4cf12dcb5c1aa",
        "Created": "2025-03-03T22:30:54.673712981+03:00",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.20.0.0/16",
                    "Gateway": "172.20.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": {
            "2e28f6c52d22b916abb96ae92663953c3c90438139e34fca8aace6f7fb04d899": {
                "Name": "database",
                "EndpointID": "649560987e88f1a829ee97d2f5874d1aa5642763edd30c2fd9824038ded28b80",
                "MacAddress": "16:e1:11:17:b5:6a",
                "IPv4Address": "172.20.0.2/16",
                "IPv6Address": ""
            },
            "fcb617c651a00becb9c4824745f0a6896888bc740048e5b8e6111a92dd762513": {
                "Name": "server",
                "EndpointID": "c79951aca9f51b3f146c57e92692b610d869b6184993ee158e6719c548a3ae95",
                "MacAddress": "e2:d8:bc:09:5b:15",
                "IPv4Address": "172.20.0.3/16",
                "IPv6Address": ""
            }
        },
        "Options": {},
        "Labels": {}
```

Создадим оверлей

```
~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp  _IGI_LR2 ?3  sudo docker swarm init
Swarm initialized: current node (juwtem6micq4d5rp8abr7wxn4) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-11894sywv6mtjhubb0r4hes6yw3zvcdcokql8i0013m27riewy-ed6mu1kklaucnt0su516oeeot 192.168.1.111:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
```

```
~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp  _IGI_LR2 ?3  sudo docker network create -d overlay overnet1
sp8dqeiae75ppvrywm4qsh0wc
~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp  _IGI_LR2 ?3  sudo docker network inspect overnet1
[
    {
        "Name": "overnet1",
        "Id": "sp8dqeiae75ppvrywm4qsh0wc",
        "Created": "2025-03-03T19:47:10.494354272Z",
        "Scope": "swarm",
        "Driver": "overlay",
        "EnableIPv4": false,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "10.0.1.0/24",
                    "Gateway": "10.0.1.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": null,
        "Options": {
            "com.docker.network.driver.overlay.vxlanid_list": "4097"
        },
        "Labels": null
    }
]
```

Создадим еще один оверлей

```
~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp  _IGI_LR2 ?3  sudo docker network create -d overlay overnet2
g4emyrmc5iif319ojafzum12z
~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp  _IGI_LR2 ?3  sudo docker network inspect overnet2
[
    {
        "Name": "overnet2",
        "Id": "g4emyrmc5iif319ojafzum12z",
        "Created": "2025-03-03T19:48:44.701499712Z",
        "Scope": "swarm",
        "Driver": "overlay",
        "EnableIPv4": false,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "10.0.2.0/24",
                    "Gateway": "10.0.2.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": null,
        "Options": {
            "com.docker.network.driver.overlay.vxlanid_list": "4098"
        },
        "Labels": null
    }
]
```

Удалим его

```
~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp  _IGI_LR2 ?3  sudo docker network ls
NETWORK ID     NAME                DRIVER    SCOPE
8c9706f6b887   Grohelbecker        bridge    local
654e177b667b   bridge              bridge    local
1b5af8d58e23   d-c_rnp_default     bridge    local
f975c64c98bf   docker_gwbridge     bridge    local
631cde74fdd9   host                host      local
exha83j2f83y   ingress             overlay   swarm
302fef477bbe   newbridgeNetwork    bridge    local
e76b2335ed9c   none                null      local
sp8dqeiae75p   overnet1            overlay   swarm
g4emyrmc5iif   overnet2            overlay   swarm
~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp  _IGI_LR2 ?3  sudo docker network rm overnet2
overnet2
~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp  _IGI_LR2 ?3  sudo docker network ls
NETWORK ID     NAME                DRIVER    SCOPE
8c9706f6b887   Grohelbecker        bridge    local
654e177b667b   bridge              bridge    local
1b5af8d58e23   d-c_rnp_default     bridge    local
f975c64c98bf   docker_gwbridge     bridge    local
631cde74fdd9   host                host      local
exha83j2f83y   ingress             overlay   swarm
302fef477bbe   newbridgeNetwork    bridge    local
e76b2335ed9c   none                null      local
sp8dqeiae75p   overnet1            overlay   swarm
```

Попробуем создать сеть host

```
~/Pr/I/353502_BURCHUCK_5/I/LR2/d-c_rnp  _IGI_LR2 ?3  sudo docker network create -d host hostnet
Error response from daemon: only one instance of "host" network is allowed
```

Докер не может создать, так как сеть типа host всего одна на машину